

PIPESENSE

IN PROGRESS

Close the supply valve

AI-NATIVE WORKFORCE SIMULATION

Practice the job before the job.

A first-person plumbing trainer controlled by your real hands, coached live by Maya, and improved after every attempt.

LEARNING LOOP

W&B • WEAVE READ

Observe — Coach — Adapt — Retry — Evaluate

Your first attempt becomes the baseline.

- Functional demo lesson: Build a P-Trap

Workspace ready. Close the supply valve, seat the trap on both pipe ends, then tighten the slip nuts.

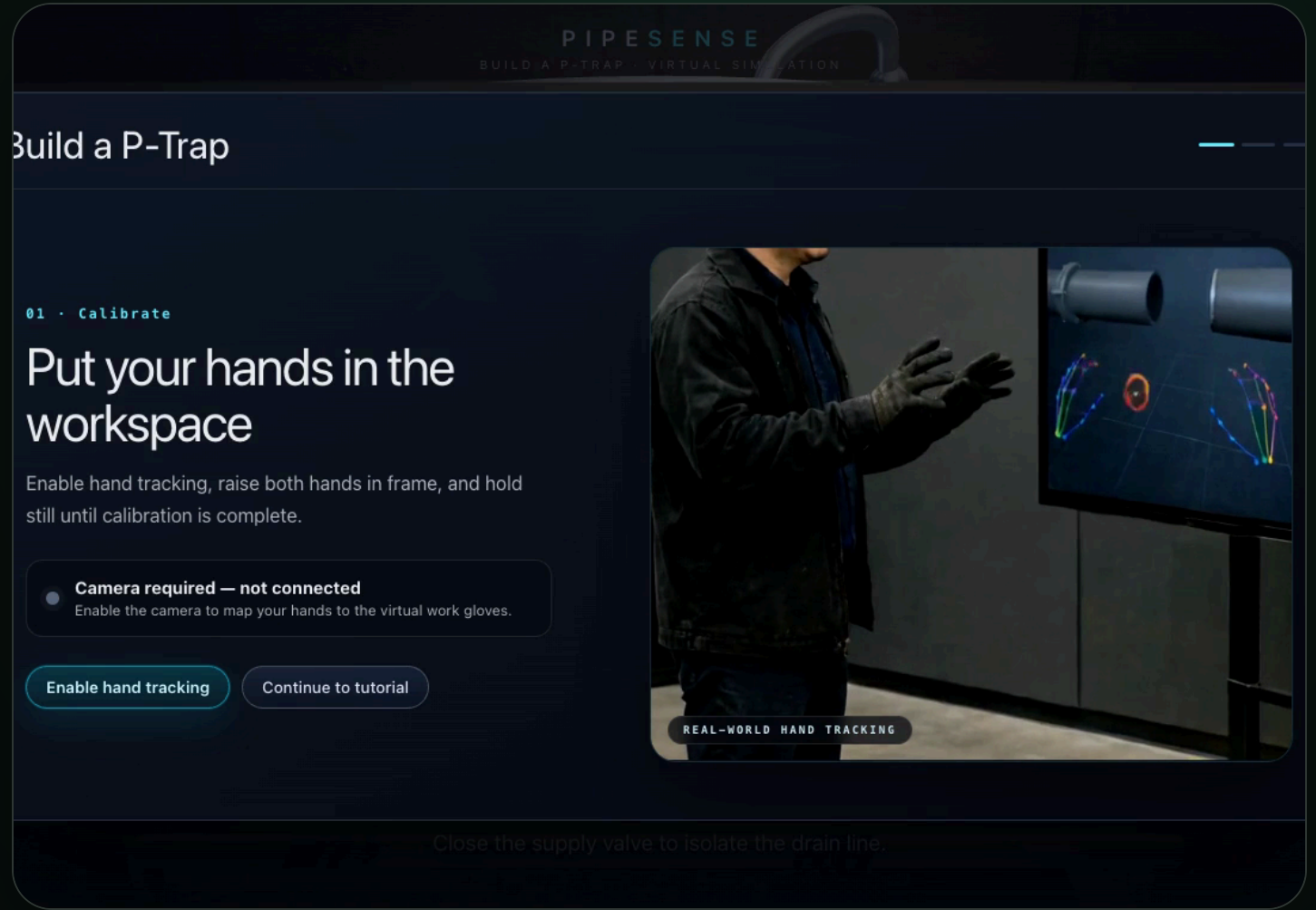
GUIDED DEMO

THE TRAINING GAP

Real repetition. Zero job-site risk.

New tradespeople need procedural confidence before a mistake reaches a customer, a pipe system, or a supervisor.

PipeSense turns webcam motion into a safe, repeatable workspace with tools, fittings, order checks, leak tests and corrective coaching.



ONE FOCUSED LESSON

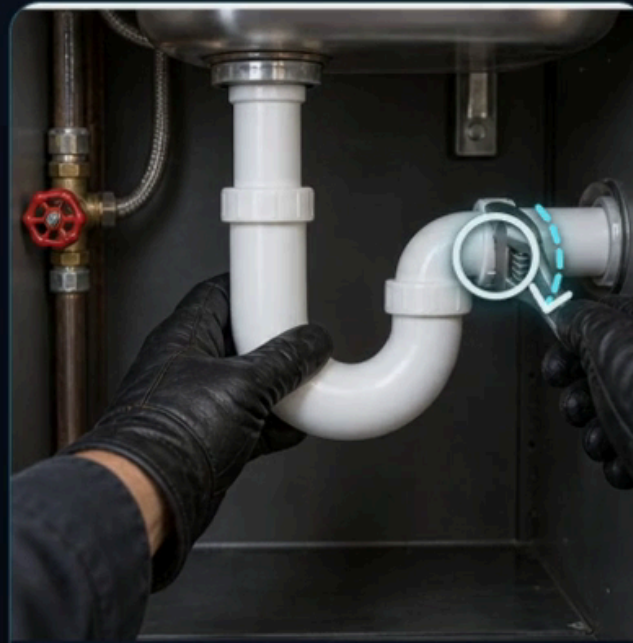
See it. Build it. Prove it.

A visual tutorial breaks the P-trap procedure into realistic, judgeable actions before the learner enters the simulation.

02 · Visual walkthrough

Learn the sequence

Review five steps, then start.

**1** Calibrate both hands**2** Learn the sequence**3** Build and test**4** Retry with coaching

WEIGHTS & BIASES IS THE LEARNING SPINE

Every attempt becomes a better next attempt.

Weave traces the learner's actions, deterministic evaluator, coach response and model inference as one inspectable attempt.

0 → 100

Observed improvement across the recorded retry loop

The screenshot displays a simulation of a plumbing task where a robotic arm is positioned over a sink. A large, semi-transparent overlay provides a detailed breakdown of the current attempt. At the top, a progress bar indicates 'STEP 05 / 05' and 'P-trap verified'. Below this, a five-step process is outlined: 1. Observe (Geometry + sequence telemetry), 2. Coach (Inference rewrites the hint), 3. Adapt (Bounded intervention chosen), 4. Retry (Simulation changes first), and 5. Evaluate (Score delta compared). The 'LATEST EVALUATED ATTEMPT' section contains a table with the following data:

SCORE DELTA	ISSUE	INTERVENTION - VERDICT	WEIGHTS & BIASES INFERENCE
100 - baseline	none / complete	reinforce (initial)	wandb-inference:openai/gpt-oss-20b-used

Below the table, the 'WEAVE TRACE' is listed as 'productmaster-nimbus/pipesense-hackathon - trace active'. The 'INPUT PROVENANCE' is 'Guided practice - simulated hands'. The 'CAMERA-CONTROL GRADE' is 'Not certifiable - simulated guided practice, no camera input'. The 'RESULT' is 'Improved - scored in browser, coached by backend'. A note states: 'Simulated guided practice. The plumbing score and any Weave trace are real; camera-control certification is never earned without a camera.' A 'Deterministic scoring' note explains that geometry, sequence, and pass/fail are decided in the browser, while the model only rewrites coaching language and the bounded next intervention. At the bottom of the overlay, it says 'Water off. WORKFLOW' and 'INTERVENTION - REINFORCE - KEEP THE CURRENT AIDS - INITIAL'. In the bottom right corner, a small inset shows a top-down view of the simulated hands with a legend: 'Guided practice - simulated hands - not ce' and 'Simulated depth - L farther z 0.50 - R far'.

Observe

MediaPipe hand landmarks

Evaluate

Geometry + procedure checks

Trace

Weights & Biases Weave attempt record

Coach

Weights & Biases Inference + Maya

Retry

Targeted practice loop

BUILT AS A WORKING PRODUCT

The sponsor tech is visible in the experience.

MediaPipe

21 landmarks per hand drive the virtual gloves and pinch interactions.

Deterministic simulation

Scores fit, sequence, leaks, retained water and the gas seal.

Weights & Biases Weave

Traces observer, coach, inference and evaluator across attempts.

Weights & Biases Inference

OpenAI gpt-oss-20b rewrites coaching; it cannot change the score.

ElevenLabs Live

Maya gives conversational voice guidance with text fallback.

Vercel

Hosts the production-ready browser demo and clean presentation mode.

BEST-FIT TRACKS

Where PipeSense competes

Best Loop Design

Best Use of Weave

Most Production-Ready

Best Social Media Demo